

STAT 444 Final Project: Option Price Surface Interpolation

Irene Chung

August 2026

1 Introduction

Option surface interpolation is a well known challenge in quantitative finance literature. Exchanges only offer a discrete number of option contracts with a finite set number of strike prices and maturity dates. For example, market data might only contain options at strikes $K = 100, 110, \dots, 200$ with maturities of 1 month, 3 months, 6 months, and 1 year. But pricing or risk management sometimes require an estimate in between those values at $K = 123$ and $T = 1.5$ months. Interpolation provides a continuous representation of the discrete market observations. This allows prices to be estimated at unobserved combinations of strike and maturity.

This is important as option prices are used for applications in option valuation, hedging, risk management, and volatility forecasting. An irregular or poorly behaved surface can produce unstable numerical estimates.

Many different methods are used in academia and industry to construct option price surfaces. This includes traditional interpolation methods like linear, bilinear, and bicubic interpolation. Parametric volatility models like the SABR or stochastic volatility models like the Heston model are commonly used. These models usually require estimating and calibrating model parameters.

The purpose of this project is to compare traditional interpolation methods with regression-based interpolation methods for constructing an option price surface. This project evaluates whether regression-based approaches such as B-spline regression and smoothing spline regression can provide comparative performance to bicubic interpolation.

2 Data

This project uses sample option chain data provided by *FirstRate Data*. The chosen option chain is for the Vanguard S&P 500 ETF (Ticker: VOO) with the trading date of 2023-07-03. The data includes different strike prices, expiry dates, last trade price, bid and ask price, bid and ask implied volatility, open interest, and various greeks (delta, gamma, etc...) for both call and put options.

Strikes ranging from 260 - 330, associated expiry dates, and bid and ask prices only for call options are used for this project.

2.1 Data Processing

Time to maturity is calculated by the difference in days of the trading date to expiry date. Strikes ranging from 260 - 330 in increments of 10 have associated maturities of 18, 46, 74, 109, 165, 200, 354, and 564 days, making a total of $8 \times 8 = 64$ different call options. We use the midpoint price between the bid and ask price,

$$mid = \frac{ask - bid}{2}$$

as the assumed call option price. The implied volatility is then calculated for each option by the Black-Scholes model using the root-finding algorithm Brent's method. We use the associated dividend rate of the VOO on July 2023 $q = 1.54\%$. The interest rate r used for each option is the closets US treasury yield rate of 1, 3, 6 months, 1 or 2 years to the options maturity in July 2023. In cases where no implied volatility can be found for the midpoint price, the ask price is used instead.

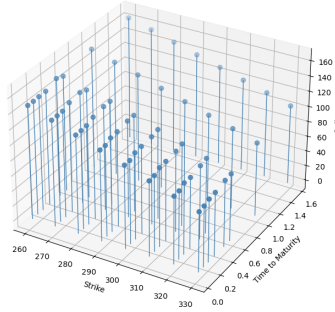


Figure 1: Call option price grid from data set.

2.2 Features Used

For implied volatility interpolation, we use the explanatory variables (or features) of strike price (K) and time to maturity (T). Our response variable we are interpolating is the call option price, C . We use the call option price instead of directly using implied volatility as call option price is needed to calculate arbitrage violations.

We interpolated over the respective grid of strikes with increments of 0.1. This gave us 700 values for of interpolation grid for both strikes and maturities. This gives us $(700) \times (700) = 490000$ values to interpolate.

3 Methodology

3.1 Models

This project compares 3 different regression models used for interpolation: Ordinary B-Splines, Penalized Splines (P-Splines), and Smoothing Splines. We also use Bicubic Interpolation as the benchmark method to compare our regression models to. We use polynomial *degree* = 3 for all models. We denote d as the number of knots and p as the number of parameters in each model.

3.1.1 Bicubic Interpolation

Bicubic interpolation is similar to a local polynomial regression method. For a grid cell bounded by (K_i, K_{i+1}) and (T_j, T_{j+1}) , coordinates are normalized,

$$u = \frac{K - K_i}{K_{i+1} - K_i}, \quad v = \frac{T - T_j}{T_{j+1} - T_j},$$

where $u, v \in [0, 1]$. The interpolated option price within the cell is represented by a bicubic polynomial,

$$\sum_{m=0}^3 \sum_{n=0}^3 a_{mn} u^m v^n,$$

which contains 16 coefficients. These coefficients are determined using the option price values and first-order partial derivatives with respect to strike and maturity, as well as the mixed partial derivative, at the four corners of the grid cell.

3.1.2 Ordinary B-Spline

Ordinary B-spline regression is

$$f(K, T) = \sum_{j=1}^p \sum_{k=1}^q \beta_{jk} B_j(K) B_k(T)$$

where

- $B_j(K)$, for $j = 1, \dots, p$, is the p B-spline basis function defined over strikes K
- $B_k(T)$, for $k = 1, \dots, q$, is the q B-spline basis function defined over maturities T

- $B_j(K)B_k(T)$ is the tensor product basis function. There are $p \times q$ possible products, forming a 2D basis over the (K, T) grid.

We denote the tensor product basis matrix as B , and the fitted beta can be obtained using B as the X matrix in OLS regression.

3.1.3 P-Splines

Penalized B-splines, better known as P-splines was first introduced by Eilers & Marx (1996). They define the second-order difference operator as

$$\Delta^2 \beta_j = \beta_j - 2\beta_{j-1} + \beta_{j-2}$$

and solve for the optimization problem

$$\underset{\beta}{\operatorname{argmin}} \|\mathbf{y} - B\beta\|^2 + \lambda \sum_j (\Delta^2 \beta_j)^2$$

The second-order difference operator can be expressed as a quadratic form

$$\sum_j (\Delta^2 \beta_j)^2 = \beta^\top D_2^\top D_2 \beta = \beta^\top P \beta, \quad P = D_2^\top D_2$$

where D_2 is the second order difference matrix. Solving the optimization problem gives

$$\hat{\beta} = (B^\top B + \lambda P)^{-1} B^\top y$$

3.1.4 Smoothing Splines (S-Splines)

We do smoothing spline regression with smoothness penalties for both strikes K and maturities T. Taking the second derivative with respect to K we get

$$\frac{\partial^2 f}{\partial K^2} = \sum_{j,k} \beta_{jk} B_j''(K) B_k(T)$$

This would give us the smoothness penalty

$$P_K = \iint \left(\frac{\partial^2 f}{\partial K^2} \right) dK dT = \sum_{j,k,j',k'} \beta_{jk} \beta_{j'k'} \left(\int B_j''(K) B_{j'}''(K) dK \right) \left(\int B_k(T) B_{k'}(T) dT \right)$$

We would also get a similar penalty for the second derivative of T. Thus, we denote the two penalty matrices P_K and P_T with Kronecker Product:

$$P_K = S_K \otimes M_T, \quad P_T = S_T \otimes M_K$$

Where S and M are defined as

$$S_K[j, j'] = \int B_j''(K) B_{j'}''(K) dK, \quad M_T[k, k'] = \int B_k(T) B_{k'}(T) dT$$

$$M_K[j, j'] = \int B_j(K) B_{j'}(K) dK, \quad S_T[k, k'] = \int B_k''(T) B_{k'}''(T) dT$$

This would give us the least squares problem

$$\|y - B\beta\|^2 + \lambda_K \iint \left(\frac{\partial^2 f}{\partial K^2} \right) dK dT + \lambda_T \iint \left(\frac{\partial^2 f}{\partial T^2} \right) dK dT$$

In which solving for this in penalty matrix form would give us

$$\hat{\beta} = (B^\top B + \lambda_K P_K + \lambda_T P_T)^{-1} B^\top y$$

3.2 Arbitrage Violations

We will check for 2 well known arbitrage conditions: butterfly spread and calendar spread. Denote the call option price as C . Option surfaces are supposed to fulfill positive butterfly spread and calendar spread

$$\frac{\partial^2 C}{\partial K^2} \geq 0, \quad \frac{\partial C}{\partial T} \geq 0$$

If your price surface violates these conditions then your surface is not arbitrage free. We approximate these values by finite difference.

$$\frac{\partial^2 C(K, T)}{\partial K^2} \approx \frac{C(K_{i+1}, T_j) - 2C(K_i, T_j) + C(K_{i-1}, T_j)}{(\Delta K)^2}. \quad (1)$$

$$\frac{\partial C(K, T)}{\partial T} \approx \frac{C(K_i, T_{j+1}) - C(K_i, T_j)}{\Delta T}. \quad (2)$$

3.3 Performance Metrics

We use 8 metrics to judge the performance of the regression models: mean absolute error (MAE), R^2 , mean relative error (MRE), number of butterfly and calendar arbitrage violations, proportion of butterfly and calendar violations, the max value of butterfly and calendar violations, and the smoothness of the strike K and maturity T direction. Smoothness in the K and T directions is defined as

$$S_K = \iint \left(\frac{\partial^2 C(K, T)}{\partial K^2} \right)^2 dK dT, \quad S_T = \iint \left(\frac{\partial^2 C(K, T)}{\partial T^2} \right)^2 dK dT$$

And is approximated through finite difference similar to section 3.2. Lower S_K, S_T implies smoother price surface.

4 Results

4.1 Fitting The Models

For ordinary B-spline, we found the optimal number of knots through Leave-One-Out cross validation. A similar process was done for finding the optimal

number of knots for the P-spline model. For the smoothing spline model, we defined a knot at every K and T value, giving us $9 \times 9 = 81$ knots. Finding the optimal tuning parameters for P-spline λ and smoothing spline λ_K, λ_T were done through General Cross Validation (GCV) error.

Table 1: Number of Knots for Each Model

Model	Knots for K	Knots for T
B-Spline	2	7
P-Spline	7	9
S-Spline	9	9

Table 2: Optimal Lambdas with GCV and Effective Degrees of Freedom

	P-Spline λ	S-Spline λ_K	S-Spline λ_T
Optimal λ	2.6473e-04	1.3412	7.5717e-06
GCV Error	1.0958	5.8614	...
df	56.75	52.9980	...

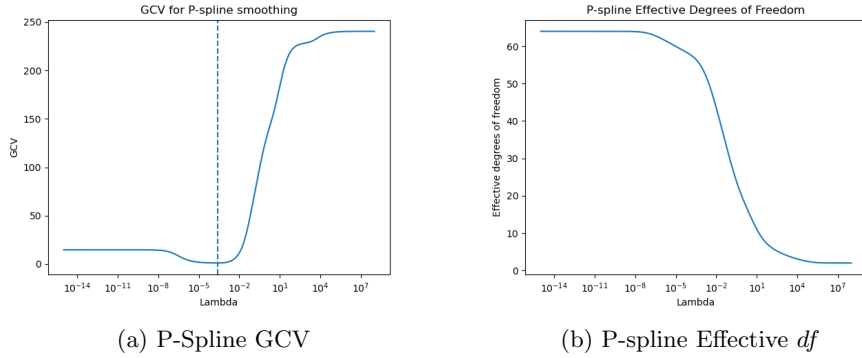
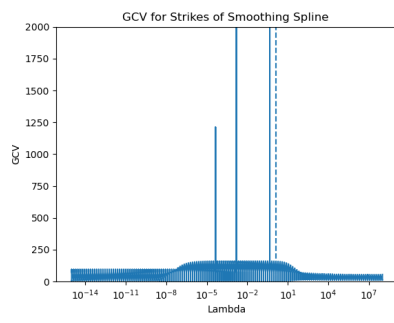


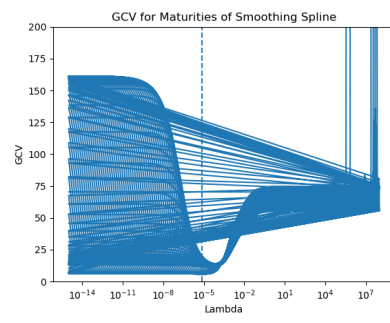
Figure 2: GCV and Effective Degrees of Freedom Graphs for P-Spline Fitting.

4.2 Model Performance Metrics

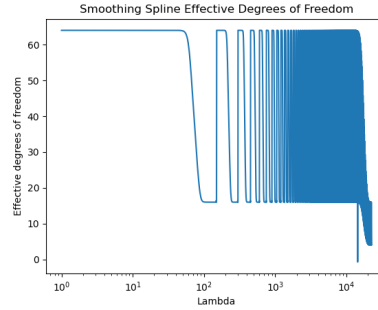
The bicubic interpolation overall performed the best out of the 4 models, with having the least amount of calendar arbitrage violations and similar butterfly arbitrage violations compared to the other 3 models. It was also overall the smoothest in both K and T directions. We do not need to look at MAE, R^2 , or MRE as the bicubic interpolation perfectly goes through each original point. So these metrics have no significance.



(a) S-Spline GCV for Strikes



(b) S-Spline GCV for Maturities



(c) S-Spline df

Figure 3: GCV and Effective Degrees of Freedom Graphs for S-Spline Fitting.

Each regression model interpolated the original 64 different option strikes and maturities very accurately, all having near 0 MAE and MRE and having $R^2 \approx 1$. In terms of arbitrage violations, the B-spline and P-spline performed the worst, with noticeably more arbitrage violations than the other 2 models. B-splines had more butterfly arbitrage violations while P-splines had more calendar arbitrage violations. This could be caused by the lower number of knots used for the respective models in the K and T axis.

Smoothing spline regression performed the best out of the 3 regression models, having a lower number of butterfly arbitrage violations than the bicubic interpolation while still having a low number of calendar arbitrage violations compared to the other 2 regression models. The smoothing spline model also had significantly smaller maximum calendar arbitrage violations than all the other models.

Visually, we can see that the bicubic surface looks the smoothest out of the 4 fitted surfaces. We can also see the curvature in the T direction on the smoothing spline surface near the bottom. The surface with the most curvature is the P-spline surface, having the highest smoothness metric in both the K and T directions.

Code and data can be found on the GitHub repository: STAT 444 Final Project: Option Price Surface Interpolation.

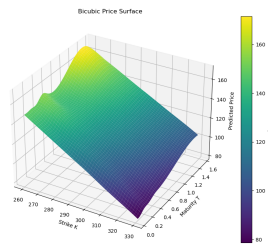
Table 3: Model Performance Comparison

Model	MAE	R^2	MRE	N_{BV}	N_{CV}	P_{BV}	P_{CV}
Bicubic	0	1.00	0	256895	64662	0.5242	0.1319
B-Spline	0.3576	0.9995	0.0030	337439	94154	0.6886	0.1921
P-Spline	0.0826	0.9999	0.0006	267547	140226	0.5460	0.2861
S-Spline	0.3375	0.9996	0.0029	256865	97549	0.5242	0.1990
	$\max(BV)$	$\max(CV)$	K Smooth	T Smooth			
	6.0595e-09	0.000332	0.0019	119248.79			
	3.6610e-08	0.000506	0.2674	622321.86			
	2.9032e-06	0.000334	0.4568	1482073.18			
	4.8991e-07	0.000042	0.3737	1447996.39			

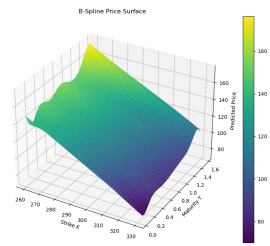
N_{BV} , N_{CV} denote number of butterfly and calendar arbitrage violations. P_{BV} , P_{CV} denotes the proportion of butterfly and calendar arbitrage violations compared to the total number of strikes and maturities. $\max(BV)$, $\max(CV)$ denotes the maximum value of butterfly and calendar arbitrage violations.

5 Future Work

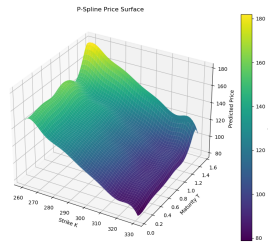
Future work could be done to thoroughly investigate the effect that more or less knots have on arbitrage violations. There are also more no-arbitrage and surface conditions like vertical spread and price bounds that could be implemented



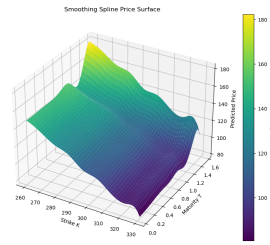
(a) Bicubic Surface



(b) B-Spline Surface



(c) P-Spline Surface



(d) S-Spline Surface

Figure 4: Fitted option price surfaces for each model.

and tested. Finally, more thorough comparisons with other industry standard interpolation methods and parametric models could be done to identify where regression-based interpolation could be beneficial over other methods.

References

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